



Experimental investigation on dominant waves in upward air-water two-phase flow in churn and annular regime



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ABSTRACT

This paper describes new measurements on the characteristics of waves in churn and annular flow. The measurements were for air-water flow in an 11 mm diameter vertical tube, a diameter close to the equivalent diameter of nuclear reactor channels. These measurements extend the parameter space of previous publications into the low and gas and liquid mass flow rates important for nuclear reactor fault studies. Detailed measurements of the flow characteristics are reported, including the identification of a new regime we designate 'pre-annular flow', in which the liquid film between the waves appears to be stationary. The results of these measurements at low mass fluxes are compared with predictions in this low mass flux region using the principal literature correlations, all of which were actually obtained from elsewhere in the parameter space. Reassuringly, the broad conclusion is that despite this, these correlations obtained from elsewhere are safe to continue to be used in this low mass flux region. A new interpretation of the criteria for onset of entrainment was also obtained from the study.

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1. Introduction

Annular flows occur in many industrial applications; boiler and evaporator tubes, heat exchangers, Boiling Water Reactors (BWRs) and so on. This flow regime is characterised by the presence of a thin wall-adhering liquid film and a central gas core. Depending on the gas and liquid velocities, the gas core may or may not entrain liquid droplets. The liquid film surface is normally wavy, the most important waves being intermittent large waves known as disturbance waves. In upward annular flow, the waves on the interface are travelling upwards. Another important regime (occurring at somewhat lower gas velocities than annular flow) is churn flow. As was seen by Hewitt et al. [1], the churn flow regime is also characterised by a liquid film at the wall and droplet entrainment in the core of the flow. However, churn flow differs from annular flow in that the liquid film reverses direction and flows downwards between the upwards-travelling large waves. Prediction of the liquid entrainment fraction and the rates of entrainment and deposition in churn and annular flow are important for mass transfer and phase change studies, and such predictions are used widely in

predicting dryout (i.e. the condition in which the film flow rate becomes zero as a result of entrainment and evaporation (Hewitt and Govan [2])).

The behaviour of waves is one of the most important characteristics in churn and annular flow. Studies of the formation and behaviour of waves in churn flow are reported by Govan et al. [3], Barbosa et al. [4] and Wang et al. [5]. Such waves are crucial in determining entrainment and onset of annular flow. There is relatively greater number of studies on disturbance waves in annular flows. These waves affect the entrainment, and possibly deposition, rates. Traditionally, most of the efforts towards prediction of entrainment and deposition are purely observational in nature, but the thrust of much current work is for these empirical approaches to become more mechanistic, informed by an (at least) qualitative understanding of the flow.

Such phenomenological modelling depends on good experimental data; on measurements of quantities such as entrainment rates and deposition rates, and on observations of 'intermediate' quantities such as the frequencies and form of disturbance waves.

There is a wide parameter space of interest, ranging from the high pressures and flows associated with normal operation in a BWR, to the low pressures and low flows important in studying various fault conditions.

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A major motivation of this paper was to augment the rather sparse availability of experimental data for nuclear-relevant small tubes, in particular by extending it into the low liquid and gas flow region, of particular interest for fault studies. This had two main objectives:

- (1) The first objective was to allow the validity of existing correlations to be investigated for this region, as these are largely based on measurements outside this region, or for which the experimental range cannot be ascertained. Demonstration that correlations used in safety analyses are truly valid for the parameter space in which they are applied is naturally necessary to satisfy regulators.
- (2) Secondly, by measuring the fullest possible set of parameters, in particular wavelength, frequency and wave speed, we hoped to provide additional insights into the flow characteristics that would aid those attempting to build semi-empirical phenomenological models of the dryout process. Disturbance waves, being large amplitude waves, have been identified as the major source of entrainment [6,7]. This fact is also highlighted by Sawant et al. [8] who have indicated the importance of disturbance wave studies for the evaluation of entrained fraction. In the literature are reports of attempts to model the rate of entrainment through simplified representations of the wave [9]. Such models require various wave characteristics such as wavelength and speed, which can only be determined experimentally.

The role of disturbance waves in influencing deposition has been less considered. Since disturbance waves change markedly the flow area available to the high-speed vapour core (e.g. Han et al. [10]), especially in small diameter channels, the core velocity will be modified by the waves, and one might expect that the rate of deposition might consequently be modified. In Fig. 1 is shown a typical disturbance wave, and an idealized sketch, with postulated vapour streamlines.

Droplets already entrained upstream of the wave, especially inertial droplets, might be deposited directly into the upstream face of the wave, if they are unable to follow the rapid radially-inwards acceleration of the vapour. The rapid radial expansion of the gas flow, and possible recirculation, downstream of the wave might similarly increase deposition rates. Since the waves themselves are believed to be a major entrainment source, it could indeed be that a large fraction of drops are born directly into this high-deposition environment. This effect of the waves on deposition rate does not feature explicitly in any of the deposition rate correlations. This might be due to the fact that normally experimental determination of deposition rate involves measuring deposition after completely removing the film and thus no waves exist when the deposition rate is measured. Thus, 'by construction', our models and correlations for deposition rate do not admit any influence of waves. On the other hand, predictions of film flow rate as a function of quality not taking direct account of the influence of waves on deposition do rather well [2]. Part of this second motivation was to provide data for a subsequent investigation into the role of disturbance waves in the deposition process.

With reference to Fig. 1, the velocity of the disturbance waves relative to the gas, as well as their size and other characteristics, all seem likely to affect the radial velocities generated. Thus disturbance wave velocity, amplitude, wavelength and time period, along with the vapour flow rate, are all important quantities that are required to be known to investigate this further. Determining these was part of our second motivation.

In Section 2 we review and discuss the relevant literature, setting the scene for our own measurements. Section 3 summarizes where the data generated in the present work lies in the parameter

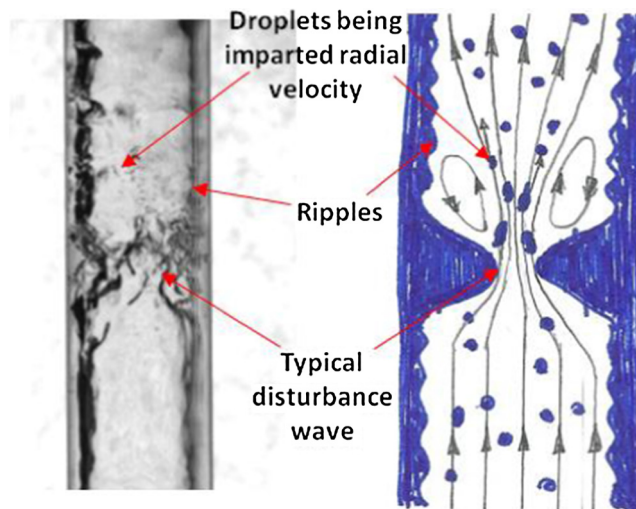


Fig. 1. Typical disturbance wave, its simplified representation, and postulated flow field.

space. In Section 4 we describe our experimental arrangements, discussing both the quantitative and the 'qualitative observations. In Section 5 we report our measurements, and relate them as appropriate to earlier literature values. In Section 6 we investigate the accuracy with which earlier correlations, predominantly based on higher flow rate conditions, predict behaviour at the low liquid and low gas flows measured. Conclusions are drawn in Section 7.

2. Literature review

Based on the mechanism of formation, in the literature large waves in annular flow are called *disturbance waves* and those in the churn flow regime are called *flooding waves*. There have been many measurements reported for disturbance waves, and only what are possibly the most relevant ones for the present work are summarized here. There are far fewer studies on flooding waves. Most of these measurements have been carried out using air and water, primarily because of the ease of experimentation.

2.1. Previous measurements on disturbance waves in annular flow regime

In the literature, pioneering and very informative work on study of disturbance wave characteristics has been done by Nedderman and Shearer [11] and Hall-Taylor et al. [12] who used cine films to determine wave characteristics in annular flow. Hall-Taylor et al. [12] used in addition conductance probes to measure the liquid film thickness, but they found that such probes tend to give spurious signals for large ripples. They found the wave velocity to be a function of both air and liquid flow rates, although the dependence on air flow rate was dominant. In both these works, wave frequency was found to be a function of both air and water flow rates. However at higher water flow rates, dependence on water flow rate was not observed [11]. The tube diameter used in these experiments was 31.8 mm, which is much larger than the ~10 mm hydraulic diameters of nuclear reactor rod bundles. Nevertheless, these experiments gave a good insight into the characteristics of disturbance waves.

Azzopardi [13] and Thwaites et al. [14] conducted similar studies on the same 31.8 mm diameter pipe, and their observations were mostly in accord with previous works. Thwaites et al. [14] found that below a particular gas flow rate the wave frequency was independent of gas flow rate, whilst above this value there was approximately linear increase with gas flow rate.

The generation of waves is related to the interplay of surface tension forces with the varying pressures induced by the vapour core flowing over a perturbed surface. One might speculate that the perturbation of the vapour flow for a given surface deformation, and the consequent tendency to induce waves, might be greater for small diameter tubes or when the surface tension is lesser. Tomida and Okazaki [15] conducted experiments in a 10 mm diameter tube. They observed that the wave frequency and wave velocity remain constant at very low gas flows but increase linearly with gas flow for a given liquid flow rate. Han et al. [10], through reconstruction of wave structures, observed that in a 9.5 mm diameter tube the area occupied by the wave was up to 20% of the total flow area. The general trends in disturbance wave characteristics reported by them were in agreement with reports for large diameter pipes. Sawant et al. [16] determined the characteristics of disturbance waves at a range of pressures above atmospheric in a 9.4 mm diameter pipe. They concluded that only two non-dimensional numbers, a modified Weber number and the liquid-phase Reynolds number, were needed to predict the dependence of disturbance wave velocity and amplitude on pressure, gas phase flow rate and liquid phase flow rate. (The Weber number is in essence a measure of the relative size of surface tension and momentum forces.)

Recently Pham et al. [17,18] have conducted visual studies on disturbance waves in air–water annular flow in a 3×3 rod bundle with spacer. Among the wave characteristics, they report measurements of the wave velocity, which they found to have a similar variation with superficial gas velocity to that seen in tubular experiments.

2.2. Previous measurements on flooding waves in churn flow regime

Compared to annular flow, experiments on flooding waves in the churn flow regime are relatively few. Notable among those are the visualization experiments of Govan et al. [3] and Barbosa et al. [4] in 31.8 mm bore tube and Wang et al. [5] in 19 and 34 mm bore tubes. Since film thickness is large in the churn flow regime, wall mounted conductance probes are prone to errors. Because of this, visualization is preferred over the conductance probe signal for determination of wave characteristics. These previous experiments were conducted with air–water at atmospheric conditions. Visualization was done at a transparent porous pipe section through which water was injected while air flowed in the core. Flooding waves were observed [3–5] to appear more frequently with increasing gas flow rate.

2.3. General observations on waves

The waves in churn and annular flow are distinctly different from each other.

Churn flow: In this regime, which obtains upstream of true annular flow, the liquid film tends to fall intermittently under the action of gravity. The downward motion of the film is decelerated when ripples are formed on the film, and these ripples are acted upon by the high speed upwards gas flow. As more ripples are generated, they tend to coalesce (as is seen, indeed, in Fig. 6 (a), from our own experiments, to be discussed later) reducing the flow area available for gas still further. Finally the increasingly obstructed gas pushes the film upwards as a large wave as observed by Hewitt et al. [1]. This type of wave is called ‘flooding wave’, or a ‘huge wave’ [5]. Often one flooding wave carries along with it many slow moving and falling ripples. This causes the disturbed region in churn flow to have a large axial extent. In this regime, it is the flooding waves which lead to a net upwards movement of liquid.

Chaotic and challenging as it is, there have recently been several attempts to make semi-quantitative models involving this churn flow region (Ahmad et al. [19]; Wang et al. [20]; Dasgupta et al. [21]). This has been motivated by the fact that having at least an approximate ‘non-arbitrary’ means to predict the initial entrained fraction is a desirable improvement to the phenomenological modelling of annular flow dryout. In turn, these efforts depend on improved observational data in this region, which was the motivation for our attempts here in this regard.

Annular flow: In this regime, both fast moving large-amplitude ‘disturbance’ waves and rather slower smaller ‘ripples’ are observed. It seems disturbance waves provide much of the impetus for the film to move upwards, generally being observed to move faster than both the ‘film itself’ and the smaller ripples. The relation between disturbance waves and ripples has been studied by Alekseenko et al. [22]. They have presented the following picture of the interaction of disturbance waves and ripples – The arrival of a disturbance wave accelerates the ripples downstream. These downstream ripples then disappear into the disturbance wave. New ripples are generated on the back slopes of the disturbance wave. These newly formed ripples initially have a velocity close to the disturbance wave but they slow down (while still having upwards motion) only to be accelerated and annihilated again by a subsequent disturbance wave.

The origin of disturbance waves itself is a subject of active research. Zhao et al. [23] have stressed upon the circumferential coherence as one of the important characteristics of disturbance waves. They have observed development of such coherence over a length of 25–50 diameters. Relatively recently, Alekseenko et al. [24] have concluded from spacio-temporal analysis of the wave streaks in downward annular flow that disturbance waves result from coalescence of ripples very near to the inlet (or beginning of annular flow) region. This is unlike churn flow regime where large waves keep getting generated by the coalescing of falling ripples far away from entry region. This is further discussed in Section 5.1.2.

3. Summary of previous measurements and present data

Table 1 summarizes some of the main relevant literature on measurement of wave properties in churn and annular flow, and the techniques used therein. For the sake of completeness, the conditions of the present experiments are also shown in the table. The present experiments aim to widen the range of measurements available in nuclear-relevant small diameter (8–12 mm hydraulic diameter) pipes, providing better coverage in particular in the low liquid and gas flow region.

The parameter space in which the present data (marked as discrete symbols) falls is shown on the Hewitt and Roberts [26] flow pattern map (Fig. 2). Also shown on the map are the parameter ranges of previous experiments [10,15,16] for determining wave characteristics in small diameter (8–12 mm) tubes. In our observations (described later in the paper), red¹ circles were of annular flow, black squares of churn, and blue triangles of the regime that we have denoted ‘pre-annular’. These observations fall very closely within the boundaries delineated in the map. It may be seen that Tomida and Okazaki [15] report experiments well into the churn and slug flow regimes but they have not discussed churn flow explicitly in their work.

The Hewitt and Roberts flow pattern map is totally experimental in nature. As against this, some flow pattern transition criteria have been developed based on transition mechanisms. Examples

¹ For interpretation of color in Figs. 2 and 3, the reader is referred to the web version of this article.

Table 1
Different techniques and measurements by researchers on waves.

References	Measurements taken						Technique used
	I.D mm	λ m	f Hz	u_w m/s	G_g kg/m ² s	G_l kg/m ² s	
Nedderman and Shearer [11]	31.8	N	Y	Y	18–103	11–231	Cine films
Hall Taylor et al. [12]	31.8	N	Y	Y	23–58	10–33	Cine films
Tomida and Okazaki [15]	10	N	Y	Y	3–100	128–910	Film thickness measurement
Azzopardi [13]	31.8	N	Y	Y	32–80	16–160	Cine films & film thickness measurement
Thwaites et al. [14]	31.8	N	Y	Y	11–41	50–379	Film thickness measurement
Wolf et al. [25]	31.8	N	Y	Y	71–154	10–120	Film thickness measurement
Han et al. [10]	9.5	N	Y	Y	100–200	18–47	Film thickness measurement
Sawant et al. [16]	9.4	Y	Y	Y	8–656	70–700	Film thickness measurement
Barbosa et al. [4]	31.8	Y	Y	Y	3–7	4–11	Cine films
Wang et al. [5]	19 & 34	N	Y	N	5–12	53–280	Cine films
Present data	11	Y	Y	Y	8–83	21–79	Cine films

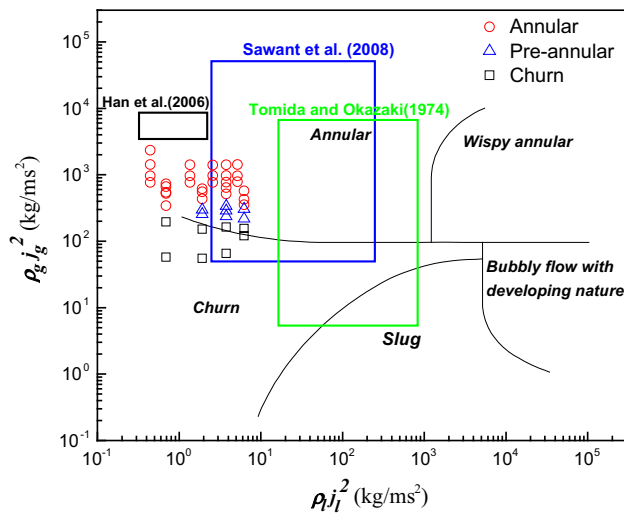


Fig. 2. The parameter space covered by the present measurements (symbols), and the ranges covered previous similar small-tube experiments, plotted on Hewitt and Roberts [26] flow pattern map.

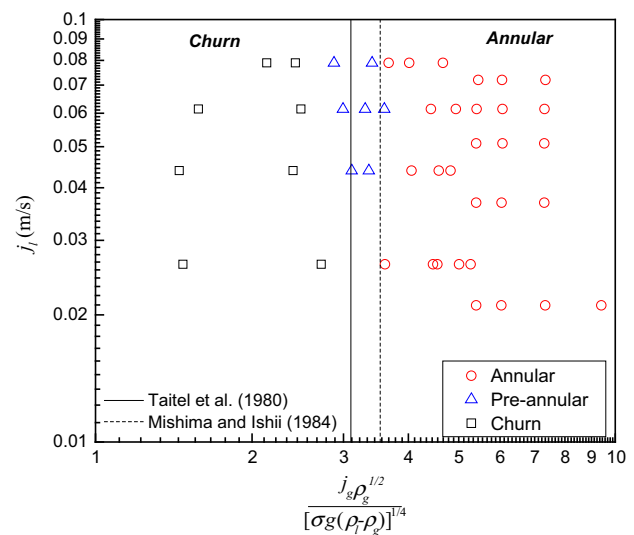


Fig. 3. Present data plotted on Taitel et al. [27] and Mishima and Ishii [28] flow pattern maps.

of such flow pattern maps are those by Taitel et al. [27] and Mishima and Ishii [28]. The present data, since it quantitatively classifies churn and annular flow regimes, provides an opportunity to evaluate these transition criteria. The evaluation is shown in Fig. 3.

It is seen that the transition criteria given by Taitel et al. [27] and Mishima and Ishii [28] are able to identify the observed transition. It may thus be said that the present data provide an additional validation for these flow pattern transition criteria.

It must be mentioned however that towards the right of the dotted line Mishima and Ishii [28] predict annular flow with entrainment. The present annular flow data (red circles) are seen to lie to the right of this dotted line. However, entrainment was not observed for all these cases. In fact, the maximum entrainment was seen to be ~6% at higher liquid and gas flow rates. The Taitel et al. [27] criterion does not make any such distinction between annular flow with and without entrainment. In these flow pattern maps also, the pre-annular flow is seen to separate churn and annular flow regimes.

4. Experimental arrangements

4.1. The facility

A schematic of the facility is shown in Fig. 4. The core of the facility is a vertical test section formed from a 2.5 m long acrylic tube, of 11 mm internal diameter. Chosen flow rates of air and

water are introduced into mixing chamber at the base of this test section. The mixing chamber (shown in Fig. 4) consists of an outer cylinder which houses within it a perforated pipe of same inner diameter as the test section. Water enters into the outer annulus of the mixing chamber and seeps in from sides of the inner pipe (having the same diameter as the test section) through perforations. The air flowing in the inner pipe carries the water along with it in the form of film at the periphery and some entrained droplets (not always!). Thus annular flow conditions can be achieved from the entry region itself. Due to stagnant tap water in the mixing chamber, brown coloured scaling is observed in the annulus region in mixing chamber.

At the upper end (~2.5 m from mixing chamber) we have two consecutive devices to extract the liquid film flowing on the wall of the tube. The extraction rate from each can be measured. At the heart of the extraction unit is a porous device across which the annular film is extracted. The inner diameter of the porous device is flush with the test section ID. Extraction rate can be varied by operating a valve (not shown here) in the 'extract out' line. Film flow rate is inferred to be that asymptotic value of liquid extraction rate at which an increase in the air extraction rate does not lead to increase in liquid extraction rate. The extraction unit additionally consists of conductance probes [29] to measure film thickness.

In the present paper, only visual (photographic) observations are reported.

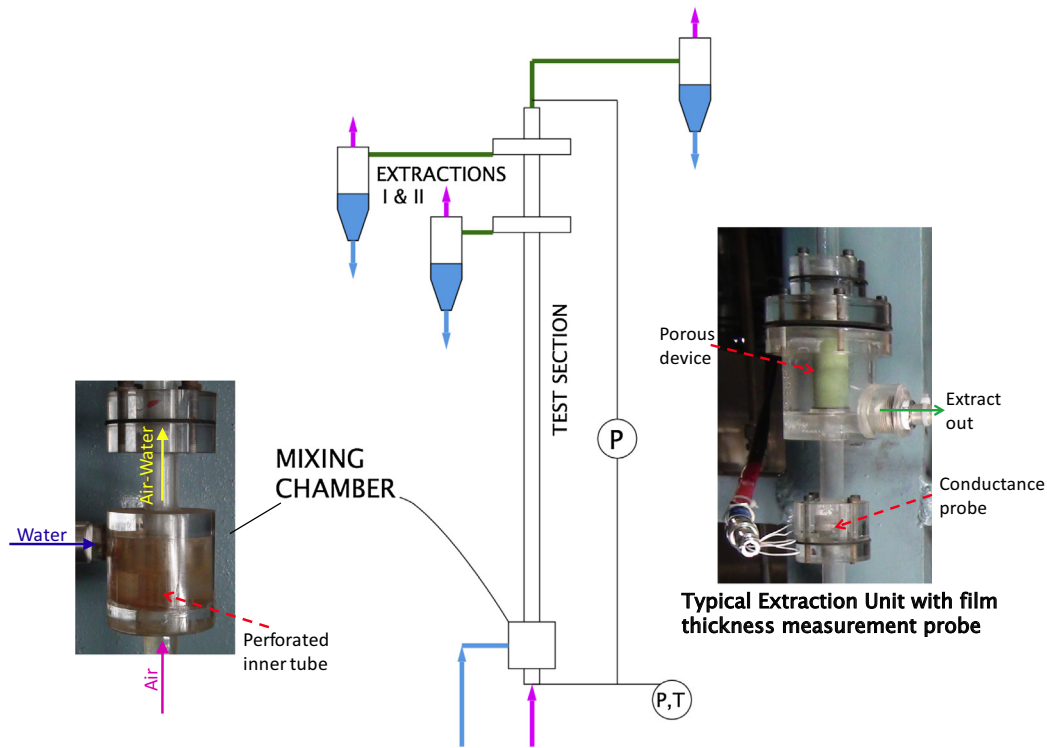


Fig. 4. The overall arrangement of the air - water annular flow facility. (Pink represents air, blue water, and green mixed.) (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

4.2. Instrumentation and analysis

4.2.1. Photographic image capture

In this work, high speed photography is used to observe and determine the characteristics of disturbance waves. With this being a transparent acrylic facility, photography can be performed at any location. Wolf et al. [25] have found that a length of ~ 100 diameters is sufficient for the development of disturbance waves. In the present setup, photography has been done approximately 2 m (182 diameters) downstream of the mixing chamber. For the churn flow regime, this location of photography is different from that used by previous workers [3–5] who had viewed the transparent mixing chamber. In their studies, they had also observed that some portion of liquid was flowing downwards from the mixing chamber. In the present study, only cases with total liquid up flow at mixing chamber were experimented upon.

A Motion Blitz camera (resolution 1 Mega pixel, adjustable x-y distribution, with a frame rate of 1000 frames per second) was used to photograph the flow in the tube. Lighting was arranged to be from behind a white screen; see Fig. 5. This provided diffuse illumination, ensuring no reflections, and allowed us to gain high resolution images of the structures in annular flow.

4.2.2. Image processing

Without image processing the natural technique to determine the wave velocity is to follow a wave manually in a high speed sequence of photographs, and measure the distance travelled by it in a particular time. This technique is however cumbersome for large numbers of waves, and is difficult to repeat and check. It is much easier to find wave velocities if wave streaks are available on a distance versus time (x-t) plot.

Distance versus time plots were generated by Hall Taylor et al. [12], by arduously tracking each and every disturbance wave from cine films. Following the technique of Alekseenko et al. [30], in this work x-t plots (see for example Fig. 6) were generated using image processing.

In essence the technique involves extracting a one pixel-wide portion of an image, giving a view of the portion of the tube length which was photographed, but of only 1 pixel's worth of the tube perimeter. The image from the same pixels in the subsequent photograph, (here) 1 ms later, is then extracted, and placed "to the right" of the first strip, and so on. This generates a composite image. This is as "tall" as the length of tube surveyed, and is one pixel wide for (here) each millisecond of the period captured in the image. Any feature of the flow (for example, a wave crest) moving up the tube, thus appears as a sloping line moving from lower left to upper right in this composite image.

An example of such an image is Fig. 6. The more chaotic the location, the darker it appears. Considering image (b), about a dozen disturbance waves enter the photographed region during the 1000 ms covered by the composite image. Their speeds are similar, but an occasional faster one is seen to 'overtake' and coalesce with the one in front of it.

Smaller, less chaotic ripples are also visible in these photographs. In (a), for example, we see a ripple left behind by left most wave, slowing down and eventually beginning to fall back, only to be engulfed by next large wave.

We will discuss the behaviour further in Section 5.

4.2.3. Wave velocities

We have explained above how the photographic images are processed to obtain the position - time information for a wave. The wave velocity is then obtained straightforwardly from this, essentially by extracting the gradient of the line associated with the wave.

It is worthwhile to discuss at this juncture the definition of a large wave, especially for churn flows where certain large waves (like the rightmost one identified in Fig. 6a) do not persist for the whole 15 cm length of the present observation window. In the present work, the characteristics of only persistent large waves are observed. These waves are distinguishable by the bold dark lines in x-t plot which persist the whole length of the frame. As against

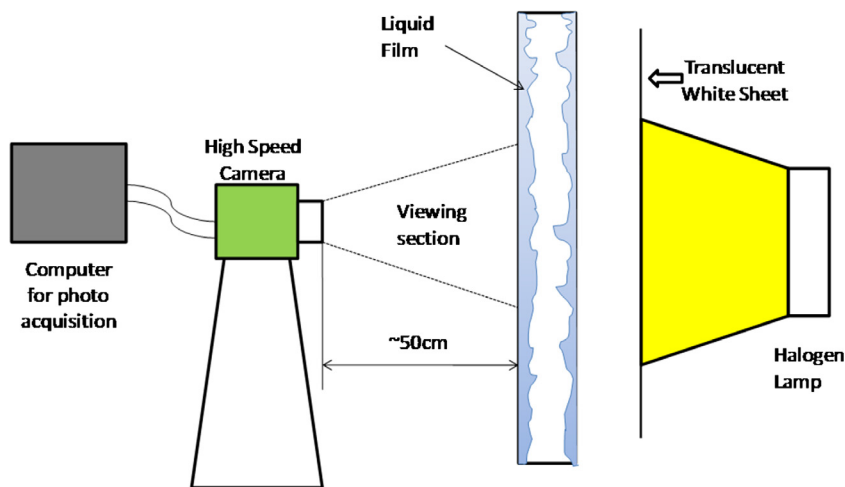


Fig. 5. Photographic arrangements.

this, the previous experiments [3–5] concentrated over a smaller axial length (of the mixing chamber) and do not distinguish between persistent and short lived waves. This might lead to the previous studies interpreting more frequent flooding waves for a given gas and liquid flow. Though short lived waves qualify to be called flooding waves, it is expected that the sustained waves would be more important in ensuring up flow of liquid than the short lived ones. Thus, the present large wave identification and characterisation seems more relevant than previous works.

4.2.4. Wave periods and wavelength

The time period, T , between successive waves was obtained from the x-t plots by measuring the distance between intersections of the wave streaks and a horizontal line on the plot. It may be seen from Fig. 6 that there can be a variation in the period between waves. The maximum, minimum and average period was recorded by observing typically 20–30 waves for each result reported.

Given then the velocities of the waves, and their period, the distance between successive waves, λ , was calculated. (Note that in the literature, and here, the term ‘wavelength’ is invariably used for this. In the present usage it refers simply to the distance between waves.)

As well as using visual observation for the above ‘quantitative’ purposes, the ability to observe high-speed images (movies) of the flow has of course allowed us to identify and report more qualitative characteristics.

5. Experimental observations

5.1. Qualitative discussion of the flow patterns observed

5.1.1. Raw flow regime images

A typical set of ‘raw’ images is shown in Fig. 7. The wave structures in churn and annular flow were seen to be very different from each other. In churn flow the film is thicker, waves are larger and so is the axial expanse of the disturbed region. The disturbed region is characterised by large upward moving waves and entrainment in the core. This is to be expected due to the intermittent counter-current motion of the liquid film and the air. This counter-current motion coincides with wave generation and entrainment from these waves. In annular flow the disturbed region occupies a much smaller length (Fig. 7). The entrainment is also thus much less than in churn flow. Based on visual observations, the extent (in terms of pipe diameters) of the disturbance region has been plotted as a function of gas momentum flux in

Fig. 8. The axial extent is taken to be from the location where increased entrainment or greater disturbance is seen on the liquid film to the point after which no such disturbances are visible. Since such a definition is subjective, especially in churn flow, the inferred extent is averaged over at least 10 waves for each value plotted in Fig. 8. For each wave in the churn flow regime, the maximum length observed was taken to be the disturbance length. The quantification of the length of the disturbed region is important in quantifying the effect of churn flow on Initial Entrainment Fraction (IEF), as has been described by Dasgupta et al. [21].

5.1.2. ‘Distance - time’ plots

The basis of these was outlined in Section 4.2.2. We will now give a commentary on the particular flow characteristics to be seen in Fig. 6.

Interesting observations can be made from these plots, amongst them being the ability to distinguish flooding waves and disturbance waves from ripples, and the ability to classify churn and annular flow *quantitatively*, from the motion of ripples.

In annular flow (Fig. 6c), both the disturbance waves and the ripples are seen to move upwards. As evidenced by the steeper slope on the x-t plots, the disturbance waves move faster than the ripples. The velocity of ripples however increases as the disturbance wave approaches and in its wake.

The churn flow regime (Fig. 6a), characterised by a falling film, is discernible as falling ripples. These ripples coalesce, forming an increasing constriction to the gas flow, till the time at which the local gas velocity at the constriction is sufficiently high to carry the falling film upwards. This gives rise to a flooding wave. At certain times the flooding waves are distinct and are carried all along the tube length. At other times, flooding waves form by ripple coalescence but do not persist all along the length. Such is evident from the case where coalescing falling ripples are shown in Fig. 6 (a). The large wave thus formed is seen not to persist. From the x-t plot, there seems to be no apparent reason for the disintegration of this wave. A possible reason could be that while the motion of this flooding wave leads to ripples emanating from its rear slope (much like the description given by Alekseenko et al. [22]), there are insufficient ripples on the downstream to sustain the large wave. In contrast, an example of a long lived large wave formed by coalescence of falling ripples is shown encircled in Fig. 9.

The behaviour in churn flow is quite distinct from annular flows where we did not observe any wave formation by ripple coalescence in the photographed zone. Large waves in churn flow are fewer (but more chaotic) than in pre-annular or annular flows.

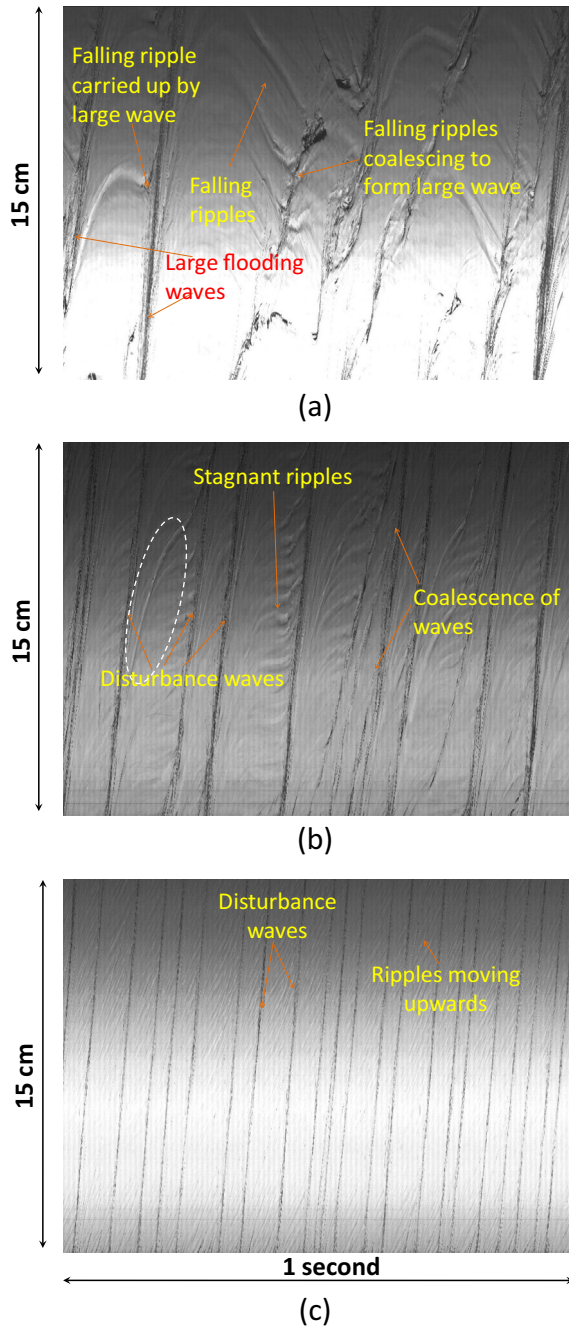


Fig. 6. Distance versus time (x-t) plots for (a) churn flow, (b) pre-annular flow and (c) annular flow.

Quite distinct from the annular and churn flow regimes where upwards and intermittently downward moving ripples are observed, a regime with momentarily stagnant ripples has been observed (Fig. 6b). We have classified this regime as *pre-annular* flow. It is also seen upon plotting on the flow pattern maps that this flow regime lies at the interface of churn and annular flow regimes. The behaviour of large waves and ripples in this regime is also intermediate between the two regimes. For instance, like churn flow, an occasional large wave (which is smaller than the usual disturbance waves) is seen to form, evidently by coalescence of ripples (Fig. 10). Such short lived low amplitude (can be inferred from the lighter streak in x-t plot, Fig. 10) waves have been identified in literature as ephemeral waves [24,31,32]. The ephemeral wave is seen to be engulfed by the next disturbance wave coming

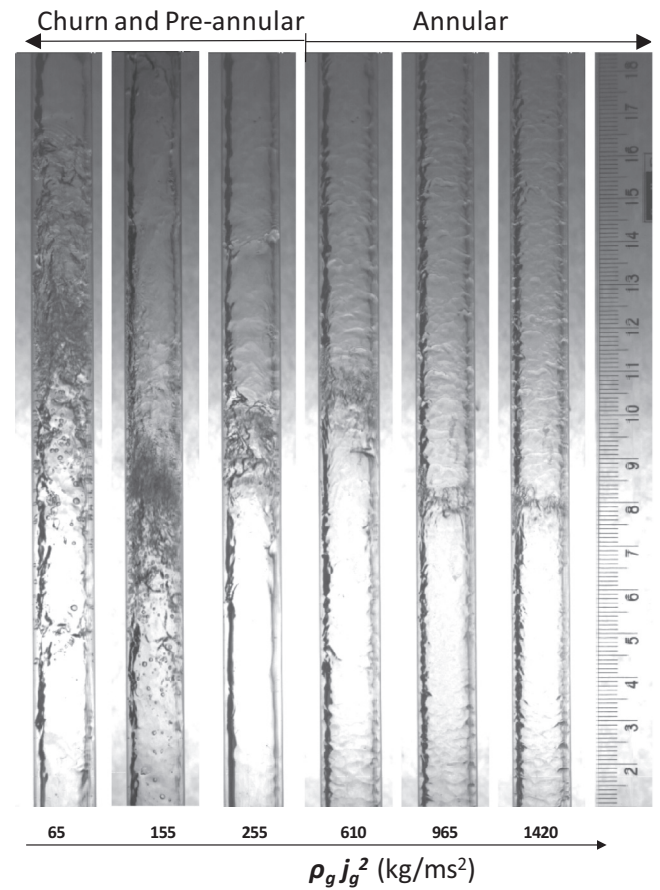


Fig. 7. Churn to annular flow regime transition as seen in terms of axial expanse of the disturbed region. Successive pictures show the increasing gas momentum flux.

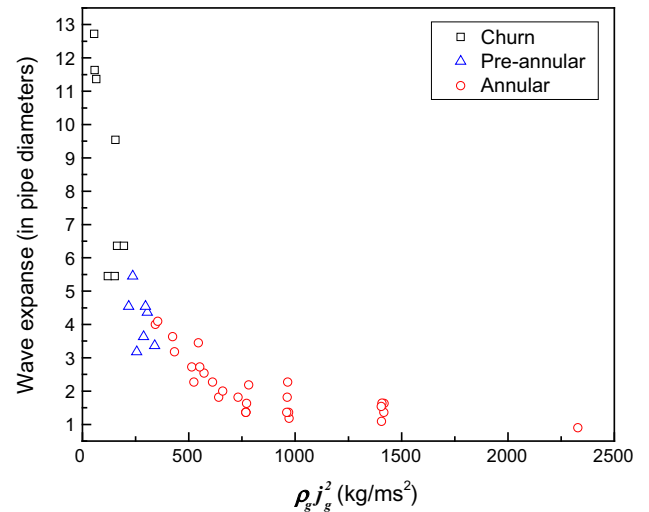


Fig. 8. Quantification of disturbance length in terms of pipe diameter.

from below. The usage of the term 'disturbance wave' (and not flooding wave) to describe the large waves in pre-annular flow follows from the fact that their parametric behaviour is similar to disturbance waves in annular flow (see Section 5.2).

Wolf et al. [25] had seen that in annular flows, fully developed wave structures are formed after about 100 diameters; wave coalescence was not observed beyond this length. In pre-annular regime, as in present case, coalescence is observed (Fig. 6b) even

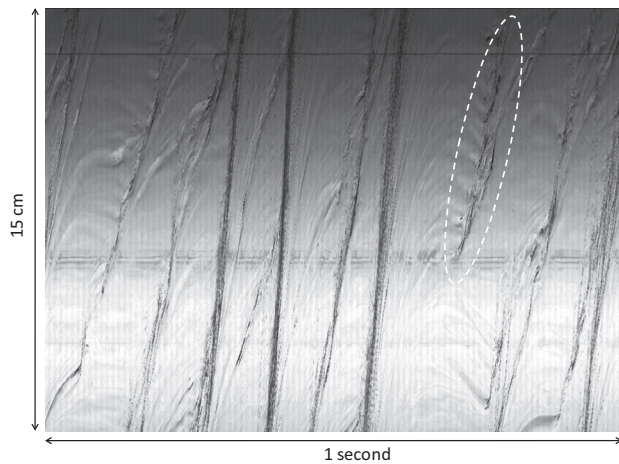


Fig. 9. Example of long lived flooding wave formed by coalescence of ripples. ($Re_l = 976$; $j_g = 9.31$ m/s).

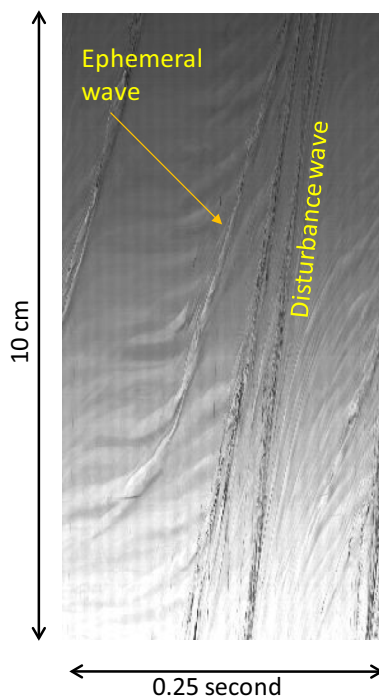


Fig. 10. Formation of ephemeral wave as seen on x-t plot corresponding to pre-annular flow regime ($Re_l = 759$, $j_g = 11.8$ m/s).

after 180 diameters downstream of inlet. It is further seen that ripples are generated not only on back slopes of disturbance waves but also on back slopes of relatively faster ripples. One such example is shown encircled in Fig. 6(b).

In view of the foregoing, it may be said that pre-annular regime identifies operating parameters in which short lived ephemeral waves are encountered along the length of the channel. Also, the wave structure is not fully developed and disturbance wave coalescence takes place at long distances from the entry.

Though pre-annular regime does not have any special significance in that it cannot be considered as a separate flow pattern and definitely is a subset of annular flow, it is expected that further study of waves in this regime would lead to better understanding of the churn-to-annular transition and whether and what exactly differentiates disturbance waves from flooding waves.

Since pre-annular regime is observed at the juncture of churn and annular flows, it will be encountered only in upwards co-current gas-liquid flow. Thus this regime is possibly of no consequence in the other flow orientations.

A general observation from Fig. 6 is the increase in the spatio-temporal uniformity of the ripples as we proceed from churn to annular flow regime. Alekseenko et al. [30] have found that the spatial frequency of ripple generation by disturbance waves in annular flow is effectively independent of gas or liquid flow rates. It may thus be inferred that a disturbance wave releases as many slow ripples as it engulfs. This, combined with our earlier inference on the absence of ripples immediately downstream of the short lived flooding wave as the reason for their short life leads us to hypothesize that the presence of ripples is necessary for sustained large wave motion. A possible picture of fully developed wave structures in annular flow is hence proposed where larger, faster disturbance waves move on a substrate of smaller, slower ripples. Further, the disturbance waves propagate by progressively accumulating the downstream ripples and releasing ripples into its upstream from its back end.

5.2. Quantitative evaluation of wave characteristics

We will show our results for wave velocity, frequency (inverse of time period) and wavelength in terms of superficial gas velocity, j_g and modified Weber number, We_m .

$$We_m = \frac{\rho_g j_g^2 D}{\sigma} \left(\frac{\rho_l - \rho_g}{\rho_g} \right)^{\frac{1}{2}} \quad (1)$$

and will note for each point also the liquid Reynolds number². It was observed by Sawant et al. [16] that when correlated against modified Weber number, the wave characteristics become independent of liquid Reynolds number.

5.2.1. Wave velocity

The wave velocities, v_w , are presented separately in Fig. 11 for churn and pre-annular flow (indicated by hollow symbols), and annular flow. In the annular flow regime the wave velocity is seen to increase with increase in j_g and We_m . For a particular superficial gas velocity or Weber number, the velocity is seen to increase with increase in Re_l . This is consistent with earlier reported experiments on annular flow. Similar dependence on Re_l is also observed in the churn and pre-annular regions.

In churn flow, however, the wave velocity reduces with an increase in gas velocity. This is in contradiction to the general presumption of increasing wave velocity with gas flow. After all, it is the gas flow which drives the film and the waves on it. The following interpretation is suggested:

In churn flow, the mechanism for upwards flow of liquid is flooding waves generated by coalescence of falling ripples. When ripples coalesce, there is initially a series of closely spaced constrictions (see for example Fig. 1, where a similar constriction is shown, albeit for annular flows), each corresponding to a falling ripple. Finally when these constrictions come close together, locally the gas velocity becomes large and the cluster of ripples is pushed up. For a given liquid throughput, reduction in superficial gas velocity would imply requirement of greater constriction in flow area to achieve flooding conditions. Greater constriction requires

² It is strictly appropriate to use two definitions of Reynolds number. The **liquid** Reynolds number we will denote and define via $Re_l = (\rho_l j_l D) / \mu_l$ where the superficial liquid velocity refers to the total liquid flow. The **liquid film** Reynolds number we denote and define via $Re_{lf} = (\rho_l j_{lf} D) / \mu_l$. In the absence of entrainment these are of course the same, and for the modest entrained fractions as in the present experiments (<6%), they differ little.

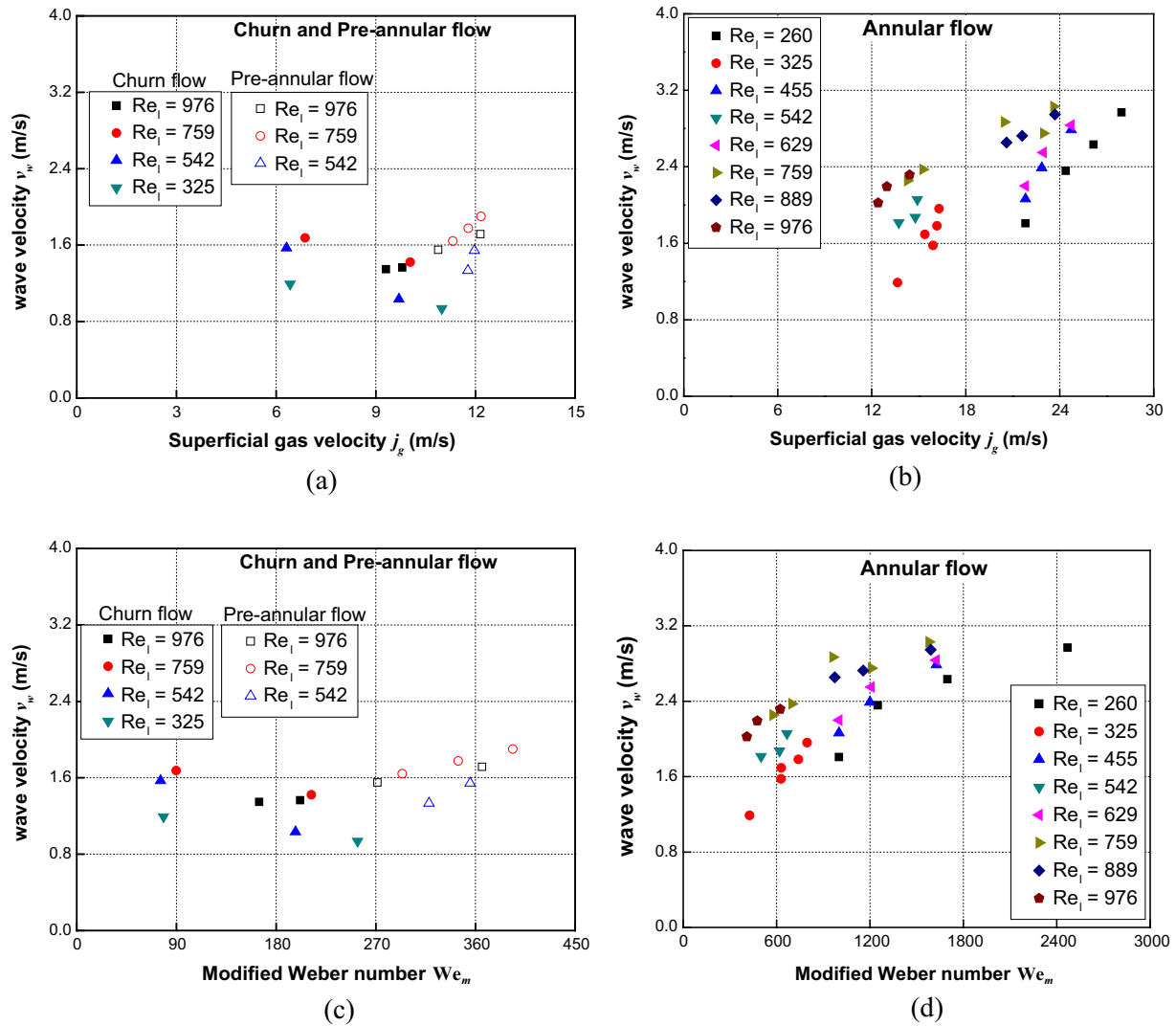


Fig. 11. Measured wave velocity versus superficial gas velocity for (a) churn, pre-annular and (b) annular flows, and versus Weber number for (c) the churn and pre-annular and (d) the annular regimes. The liquid Reynolds number for each set is indicated.

coalescence of larger number of ripples. Since the mass of liquid now required to be moved up by a single flooding wave increases, the local velocity requirement for flooding also increases. This leads to a greater constriction in flow area and a higher gas velocity at that location. The higher gas velocity would lead to greater wave velocity. Another complementary line of reasoning would be that for a given liquid flow rate (Re_l), mass balance has to ensure that the liquid throughput is maintained. If greater number of ripples coalesce, the flooding wave, which maintains the positive up-flow, has to move at higher velocity.

5.2.2. Gas-wave relative velocities

The velocity of the gas relative to the wave is one of the quantities that is expected to be important for simulations of droplet deposition phenomena. Specifically such data (along with information on wave shape) are useful while applying Lagrangian mechanics to investigate the influence of disturbance waves on deposition. In literature, though Lagrangian studies have been undertaken, the presence of waves has not been acknowledged [33,34]. Measurements of this relative velocity are one of our major motivations.

The relative velocity, $(j_g - v_w)$ is plotted in Fig. 12 for the different cases. In view of the fact that the average film thickness is small in comparison to the tube diameter (in the present experi-

ments, film thickness was seen to be of the order of 0.1 mm while the tube ID is 11 mm), $(j_g - v_w)$ can be considered to be a good approximation of the velocity of gas relative to the wave. Lines corresponding to constant wave velocity of 0 m/s and 3 m/s are also shown for clarity in Fig. 12(a) and (b).

The relative velocity increases with j_g in all the flow regimes. Though the wave velocity was seen to depend on Re_l , the relative velocity shows no observable dependence. This might be because relative velocity is much higher than wave velocity. With Weber number too, there is an increase in relative velocity in the churn and pre-annular flow regimes.

5.2.3. Wave frequency

The wave frequency (the inverse of the time period between successive waves) obtained in the present experiments is plotted in Fig. 13. As mentioned earlier and observable in distance-time plots (Fig. 6), there is a band of frequencies. Further, the maximum and minimum frequencies are not distributed evenly about the average value. The bars shown in Fig. 13 represent the observed variation in wave frequency (and not errors in the measurement).

Typical histograms of frequency distribution in churn, pre-annular and annular flow regimes are shown in Fig. 14. In all the

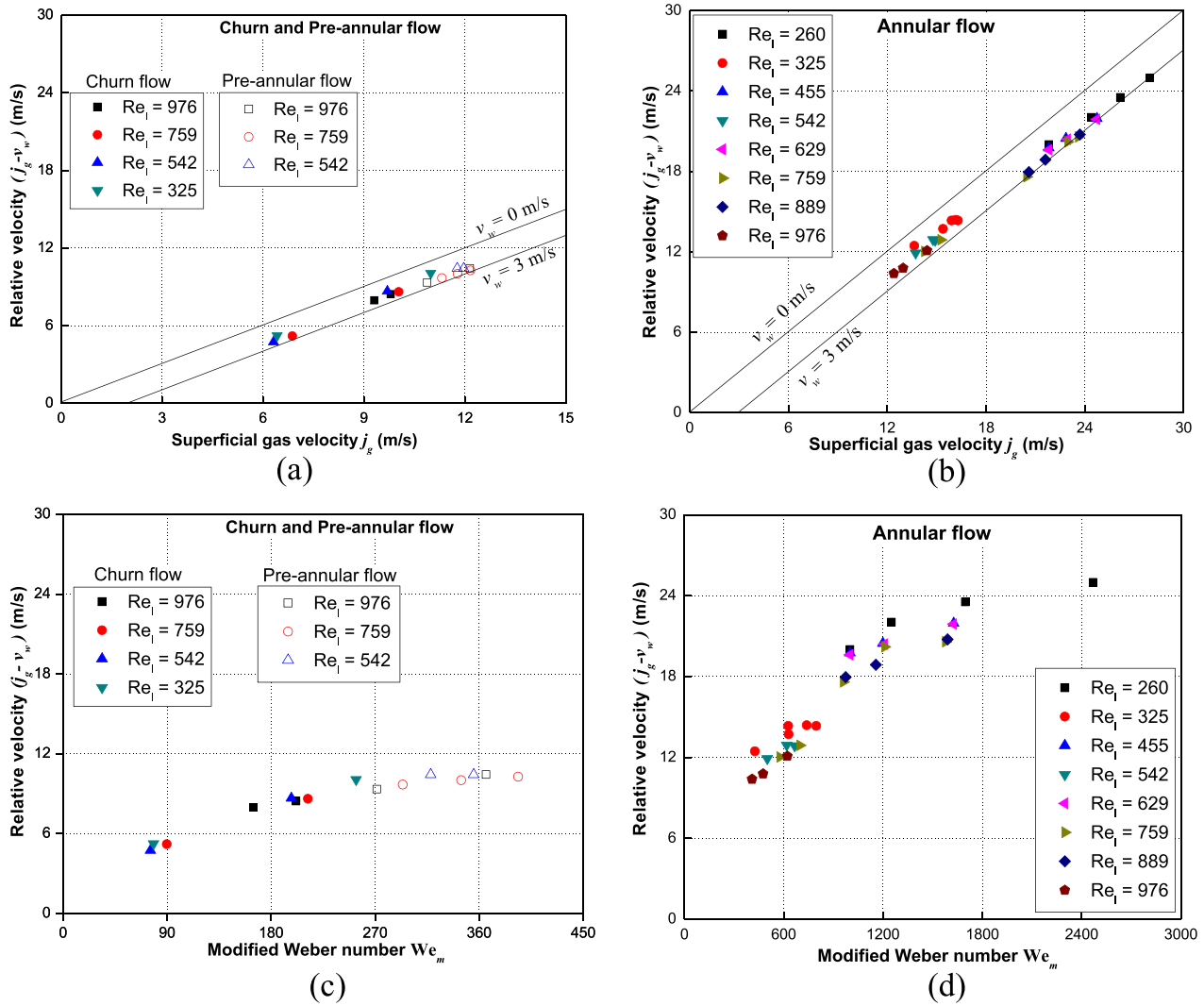


Fig. 12. Measured relative velocity versus superficial gas velocity for (a) churn, pre-annular and (b) annular flows, and versus Weber number for (c) churn, pre-annular and (d) annular flows.

cases, it is seen that there are few waves which come in very quick succession while the average frequency is quite low.

In the present study the average frequency is seen to increase both with increase in gas flow rate (and Weber number) as well as liquid flow rate in the annular flow regime. This is in line with the observations of Han et al. [35] and Nedderman and Shearer [11]. However we have not observed, in annular flow, any limiting j_g below which average frequency is independent of j_g as was observed by Webb [36], Thwaites [14] and Chu [37]. The waves are generally less frequent in the churn and pre-annular flow regimes. In these regimes, the frequency is not seen to be a strong function of either gas flow rate or liquid flow rate. But some increase is seen with gas flow rate.

5.2.4. Wavelength

The wavelength in the present context refers to the average distance between two successive large waves. In the present study it is obtained by dividing the wave velocity by the average frequency. It can be observed from Fig. 15 that average wavelength decreases with the increase in gas flow in churn flow. Thus the flooding waves become more closely spaced as gas flow increases.

In annular flow, the variation of wavelength with gas flow for a given liquid flow is not so marked. However, there is a strong

dependence on the liquid Reynolds number, Re_l particularly at lower Re_l . An interesting inference can be drawn from this observation by considering that for low entrainment in annular flow regime, as in present case, we may take the liberty of equating film and liquid Reynolds numbers i.e., $Re_{lf} \approx Re_l$. The variation of average wavelength with film Reynolds number is thus plotted in Fig. 16. It is notable that for film Reynolds number values less than about 400 the distance between disturbance waves rises sharply.

A critical film Reynolds number, Re_{lfc} for the onset of entrainment was identified by Hewitt and Govan [2] and correlated as

$$Re_{lfc} = \exp \left(5.5804 + 0.4249 \frac{\mu_g}{\mu_l} \sqrt{\frac{\rho_l}{\rho_g}} \right) \quad (2)$$

It is interesting to note that the predicted value of Re_{lfc} for our conditions is about 415–450, which is also approximately the value of film Reynolds number below which the wavelength suddenly begins to increase rapidly. Further, the film flow rate evaluation (not reported here) done for the present data points (as in Fig. 2) indicated that no entrainment was observed till $Re_l = 455$. This provides an additional validation for applicability of Eq. (2) for evaluation of the minimum film flow rate for entrainment to occur.

To investigate further the criterion for onset of entrainment, high-speed films for flows of different liquid flow rates with nearly

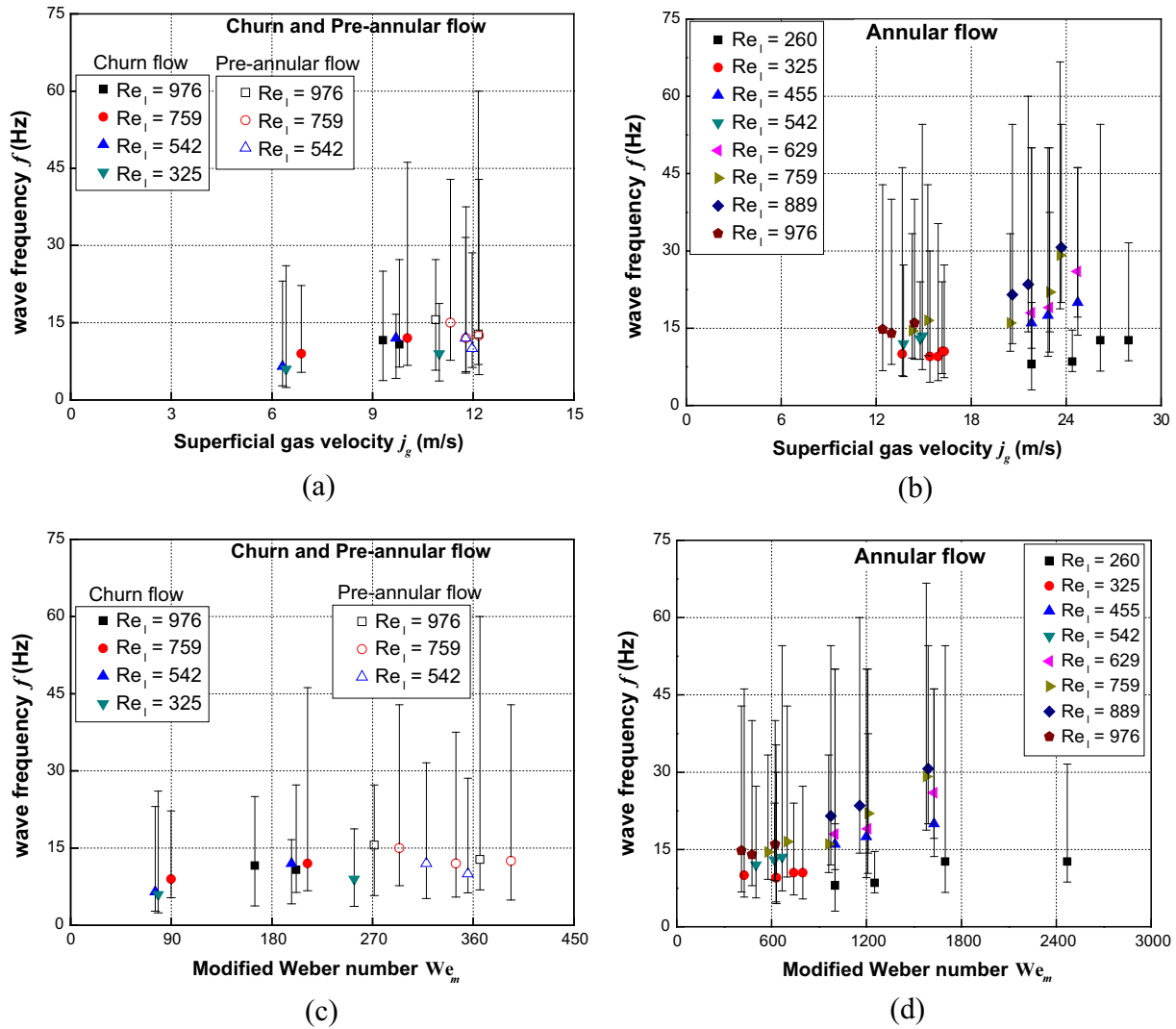


Fig. 13. Measured wave frequency versus superficial gas velocity for (a) churn, pre-annular and (b) annular flows, and versus Weber number for (c) churn, pre-annular and (d) annular flows.

the same gas flow rates were studied, as illustrated in Fig. 17. Since the present annular flow data were seen to lie in the low entrainment zone (maximum entrainment of $\sim 6\%$), we cease to make a distinction between liquid and film Re , as for low entrainments, they are essentially equal.

Some of the entrained drops are marked by arrows in Fig. 17. In this figure, disturbance waves are observed for film Reynolds numbers below Re_{ifc} and entrainment is observed from the wave tips. This is in contrast to the reasoning behind the existence of a critical film Reynolds number given by Ishii and Grolmes [7], who suggest that it is the absence of disturbance waves below Re_{ifc} that would lead to cessation of entrainment. Their Re_{ifc} is however much lower than that given by Eq. (2). Hewitt and Govan [2], however, do not discuss the physical interpretation of Re_{ifc} as given by this equation.

It is seen from Fig. 17 that though entrainment is observed from the tip of the waves for $Re_{if} < Re_{ifc}$ the entrainment does not survive downstream of the wave. The probable recirculation effect (Fig. 1) is seen for $Re_i = 325$ in Fig. 17. Beyond Re_{ifc} however, the entrained drops can be seen in videos (and to a lesser extent in the stills in Fig. 17) to survive a long distance downstream of the wave.

It is thus postulated that for sustained entrainment to exist, the entrainment from one wave must be at least carried in the gas stream up to the next downstream wave. This physical significance

of Re_{ifc} is reinforced in Fig. 16, which relates the critical film Reynolds number to wavelength.

5.3. Experimental uncertainties

For phenomena such as these and these kinds of measurements, uncertainties themselves can only be estimated fairly approximately. However, we will attempt to provide an indication.

5.3.1. Flow rates

The water flow rate was calibrated by measuring the water flow in a laboratory measuring cylinder. For air flow rate we calibrated against a standard rotameter. The inlet water and air flow rate measurements were accurate to within ± 0.02 and ± 1.7 L per minute. This translates to a maximum error of $\pm 16\%$ in water flow rate and 3.2% in air flow rates.

5.3.2. Wave velocities

The wave velocity is estimated by measuring the slope of the streaks on x-t plots. This slope was measured manually using a protractor, and the error is estimated to be around 11% .

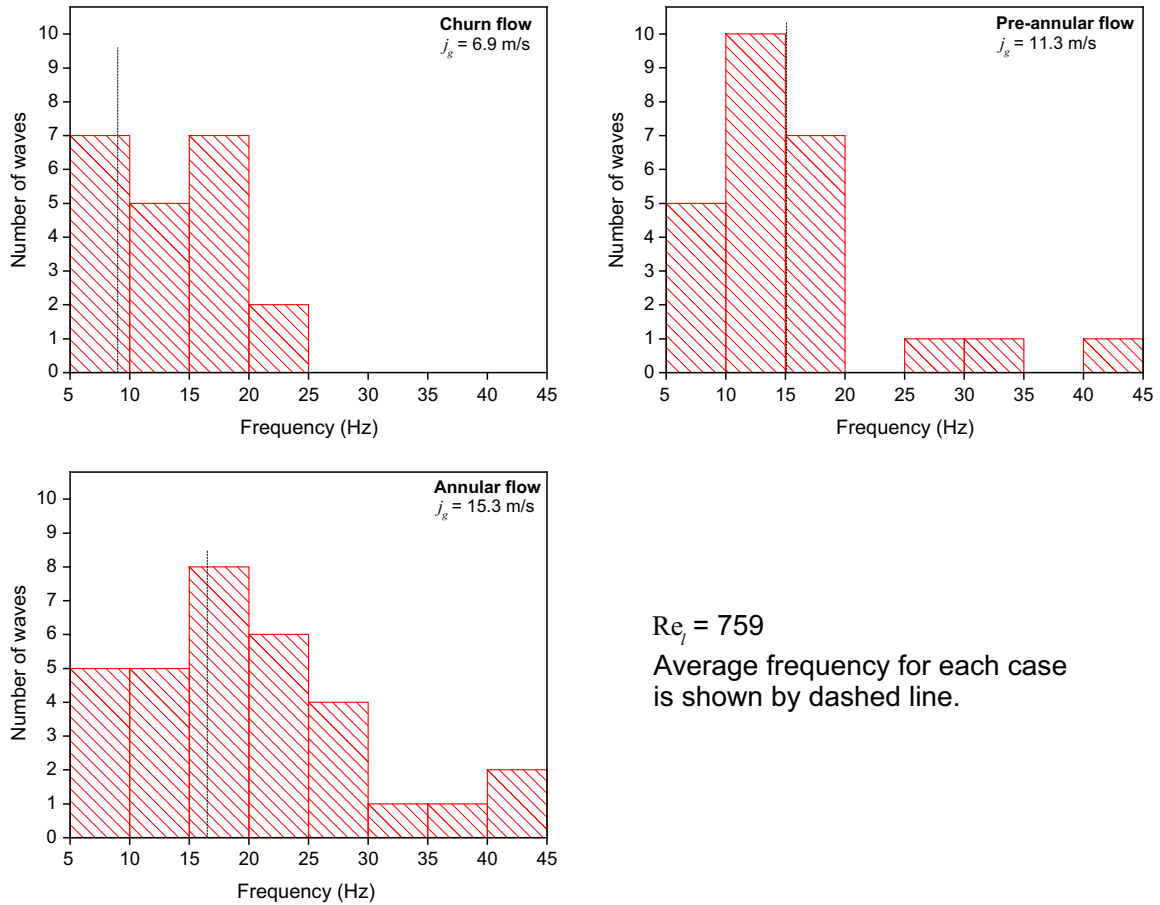


Fig. 14. Histograms of wave frequency for churn, pre-annular and annular flow regimes.

5.3.3. Axial extent of the waves

The resolution from the digitization of the images is $\sim 0.2 \text{ mm}$, which is approximately the length of 1 pixel, and lengths can thus in principle be determined with this accuracy. Whilst in annular flows the axial extent of a wave does not vary to a large extent, in churn flows there is an inherent increase and decrease in the wave expanse as the wave moves along. We have taken the highest values observed to plot the axial expanse graph (Fig. 8).

5.3.4. Wave frequencies and wavelengths

Due to the frame rate of the camera we are limited to a time resolution of 1 ms (thus an error of $\pm 0.5 \text{ ms}$). The periods between successive waves can also be measured with the same accuracy. The frequency is the inverse of time period.

$$f = \frac{1}{T}; \left| \frac{df}{f} \right| = \left| \frac{dT}{T} \right| = \frac{0.5}{15} = 0.033 \quad (3)$$

Thus, greater the time period (or lesser the frequency), lesser is the error. For high frequency the error would be larger. For our case the maximum frequency is $\sim 67 \text{ Hz}$. This implies a time period of $\sim 15 \text{ ms}$ and a max error of $\pm 3\%$.

Since wavelength is related to wave velocity and frequency, we may write:

$$\left| \frac{d\lambda}{\lambda} \right| = \left| \frac{dv_w}{v_w} \right| + \left| \frac{df}{f} \right| \quad (4)$$

Thus we may say, conservatively, that the maximum uncertainty in wavelength is $\pm 15\%$.

For the avoidance of doubt, we would just reiterate that the error bars shown on Figs. 13 and 15 do not represent errors in

measurement, but the actual variation in observed frequency and wavelength. The ranges indicated by the bars are thus a characteristic of the flow itself. Whilst this type of reporting, of maximum, minimum and average values, is not generally observed in the literature, we believe it is important.

In the foregoing, errors in deterministic procedures are analysed. However, the process of large wave generation is inherently stochastic. Thus, when the sample size is larger, the confidence level in our experiments (and the errors proclaimed) increases. This implies, for instance, in churn flow, fewer waves are observed leading to low frequency, thus though the deterministic error is smaller (Section 5.3.4), the confidence level is lower as the number of waves studied were lesser than in annular flow. Such a statistical analysis, though important, has not been performed in the present work as the sample size of large waves for the different operating points is different which would lead to variable confidence bands for the different sets.

It must however be mentioned that variation of wave velocity among the different waves for a particular air-water flow rate combination was negligible. Thus, wave velocity is not likely to be affected by the statistical uncertainty.

6. Comparison of measured wave characteristics with available correlations/data

The present experiments explore wave behaviour in a part of the parameter space little covered in earlier works, but which is important in practical applications. Correlations available to the analyst are predominantly derived from results elsewhere in this space (or for which the parameter space of their derivation is

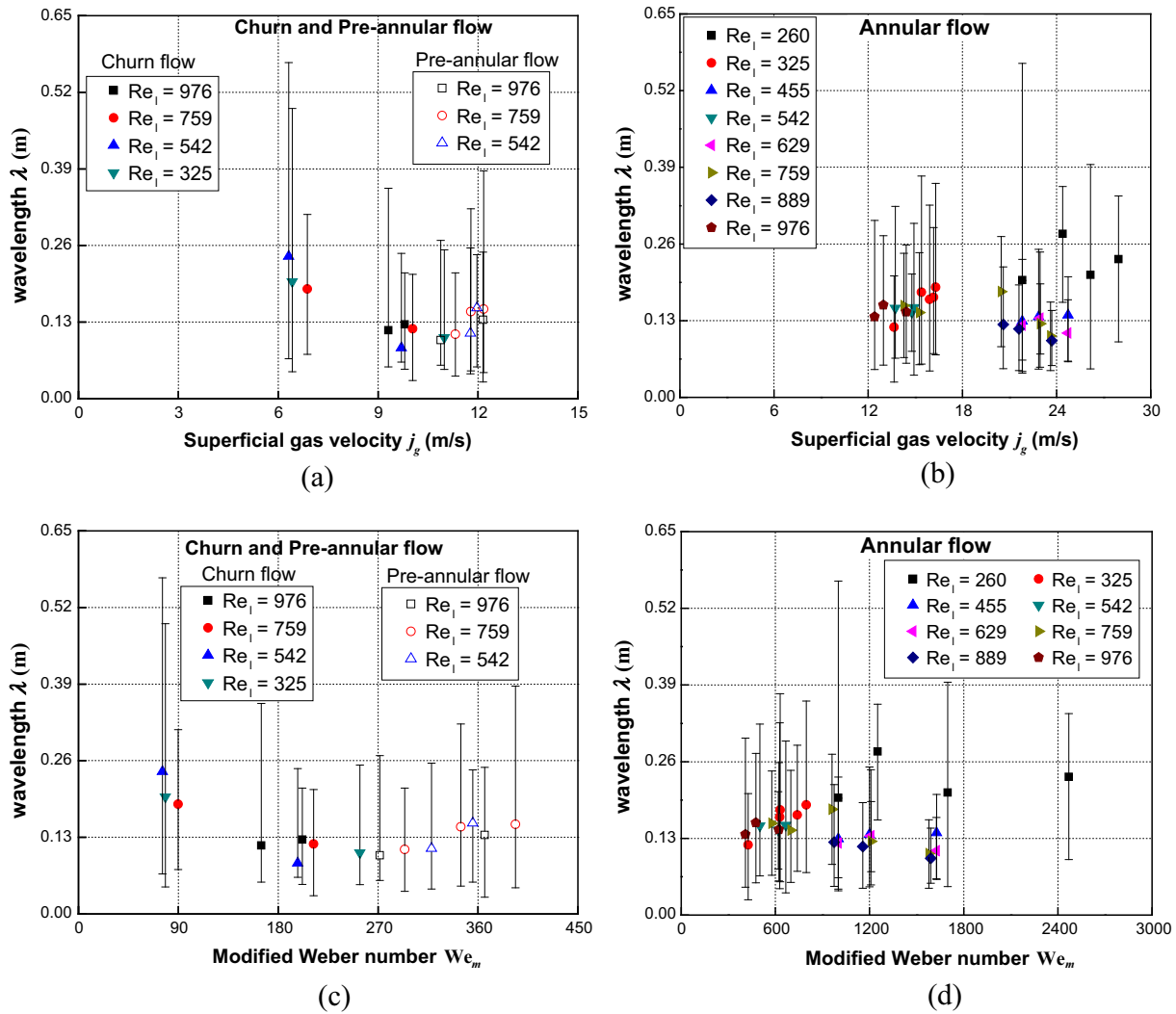


Fig. 15. Measured wavelength versus superficial gas velocity for (a) churn, pre-annular and (b) annular flows, and versus Weber number for (c) churn, pre-annular and (d) annular flows.

uncertain) and we study here their effectiveness in predicting behaviour of the low-flow space under investigation. It must be noted that the correlations considered are applicable only in the

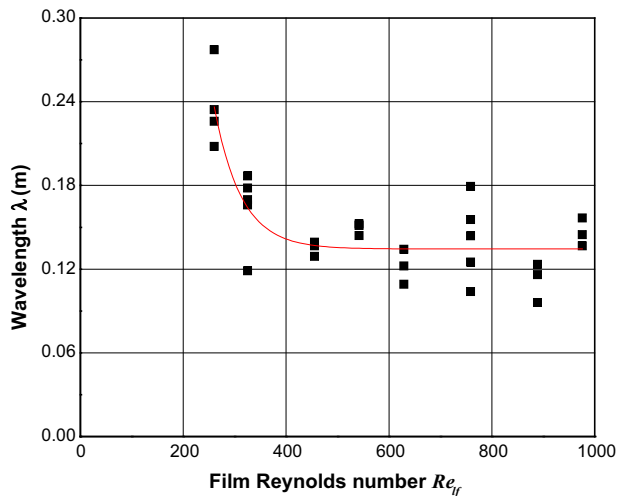


Fig. 16. Measured wavelength as a function of film Reynolds number for the annular flow regime.

annular flow regime and thus are compared only with the present annular flow data.

Though an analytical model is available [4] for the velocity of flooding waves in the churn flow regime, and a correlation is available [5] for their frequency, considering the complexity of churn flow it is considered more appropriate to compare the present velocity and frequency data against previously reported data.

6.1. Wave velocities

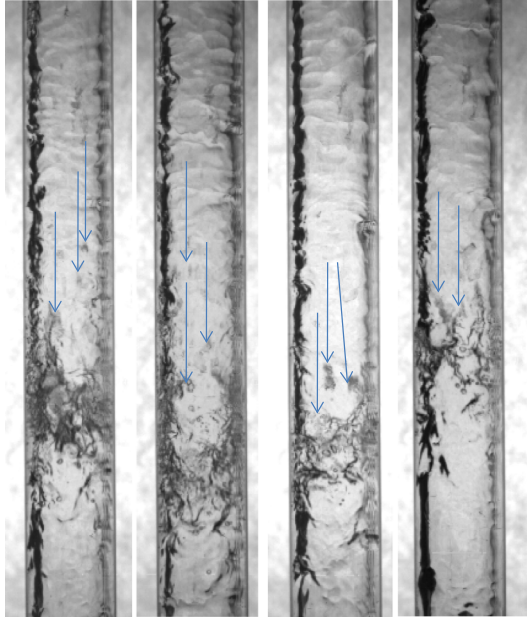
6.1.1. Pearce correlation for annular flow

The wave velocities (in the annular flow regime) obtained in this experiment are compared with predictions using Pearce [38] correlation:

$$v_w = \frac{K_{pr} \bar{u}_{lf} + j_g \sqrt{\frac{\rho_g}{\rho_l}}}{K_{pr} + \sqrt{\frac{\rho_g}{\rho_l}}} \quad (5)$$

Here $\bar{u}_{lf}$ is the time-averaged liquid film velocity, and K_{pr} is a parameter depending on pipe diameter. The time-averaged liquid film velocity is obtained from Eq. (6).

$$\bar{u}_{lf} = \frac{d_{eq} G_{lf}}{4 \rho_l \delta} \quad (6)$$



j_g (m/s)	14.4	14.2	14.8	15.4
$Re_{lf} \approx Re_l$	976	759	542	325
Re_{ffc}	418.72	422.18	422.18	426.18

Fig. 17. Measured limiting criteria for the onset of entrainment (a few entrained drops are marked by arrows).

The average film thickness, δ , was evaluated experimentally by using conductance probes [29] as mentioned earlier. As noted above, appreciable entrainment does not exist for the present experimental data set. Thus the film mass flux, G_{lf} was assumed to be equal to the inlet liquid mass flux for computation of $\bar{u}_{lf}$. For the present case, the value of K_{pr} is taken as 0.63 for our pipe diameter of 11 mm, following the prescription by Ombere-Iyari and Azzopardi [39], who conducted experiments with different pipe diameters.

The predictions by the Pearce [38] correlation are plotted in Fig. 18. It is seen that the correlation over predicts by 30% for the present data set. Sawant et al. [16] also used this correlation for predicting their data and obtained similar accuracy.

6.1.2. Kumar et al. correlation for annular flow

The Kumar et al. [40] correlation was derived from experiments in the range

$$\rho_g j_g^2 = 4700\text{--}8300 \text{ kg/ms}^2$$

$$\rho_l j_l^2 = 7.5\text{--}600 \text{ kg/ms}^2$$

which is wholly outside the space of our experiments (and indeed, was from experiments in a rectangular channel 63.5 mm $\times$ 63.4 mm). This correlation is

$$v_w = \frac{K_{kum} j_g + j_l}{1 + K_{kum}} \quad (7)$$

where K_{kum} is a constant given by Eq. (8)

$$K_{kum} = a \left(\frac{\rho_g}{\rho_l} \right)^{\frac{1}{2}} \left(\frac{Re_l}{Re_g} \right)^{\frac{1}{4}} \quad (8)$$

Since as noted this correlation, is for a rectangular channel whereas ours is circular, the constant K_{kum} is obtained with $a = 7.653$, the appropriate value for the present experiment.

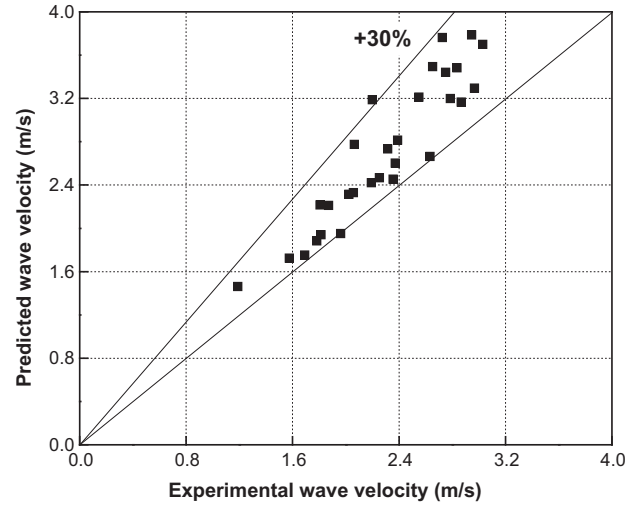


Fig. 18. Comparison of the present measurements of the wave velocity with the Pearce [38] correlation.

Using this, the predictions of the correlation are compared with our measurements in Fig. 19. As for Pearce correlation, the predictions of the correlation are reasonably consistent with the present measurements; the correlation predicts most of the present data within +20% and –10%, which is better than the Pearce [38] correlation. Sawant et al. [16] also obtained better agreement of their data with the Kumar et al. [40] correlation than the Pearce correlation.

6.1.3. Flooding wave velocities in churn flow

A comparison of the present data with the previous experimental data [13] on wave velocity and frequency is shown in Fig. 20. The wave velocities are plotted with non-dimensional superficial velocity, j_g^* .

$$j_g^* = j_g \sqrt{\frac{\rho_g}{gd(\rho_l - \rho_g)}} \quad (9)$$

In the zone of the present data ($j_g^* > 0.75$), the trend of wave velocity (i.e., reduction upon increasing gas flow) observed is mostly similar. However the wave velocity is seen to be much smaller than previous experiments for the same j_g^* and liquid mass flux. The effect of liquid mass flux is not found to be as significant as in previous works.

6.2. Wave frequencies

6.2.1. Sekoguchi correlation for annular flow

Disturbance wave frequencies have been correlated by Sekoguchi et al. [41]. The correlation is quite complex with the frequency, f correlated as Strouhal number, Sr which is a complex function of Eötvös number, Eo , Froude number, Fr_g and liquid Reynolds number Re_l .

$$Sr = \frac{fd}{j_g} = f_1(Eo)g_1(\psi) \quad (10)$$

where

$$f_1(Eo) = Eo^{-0.5} (0.5 \ln(Eo)) g_1(\psi) \quad (11)$$

and

$$g_1(\psi) = 0.0076 \ln(\psi) - 0.51 \quad (12)$$

$$Eo = \frac{gd^2(\rho_l - \rho_g)}{\sigma} \quad (13)$$

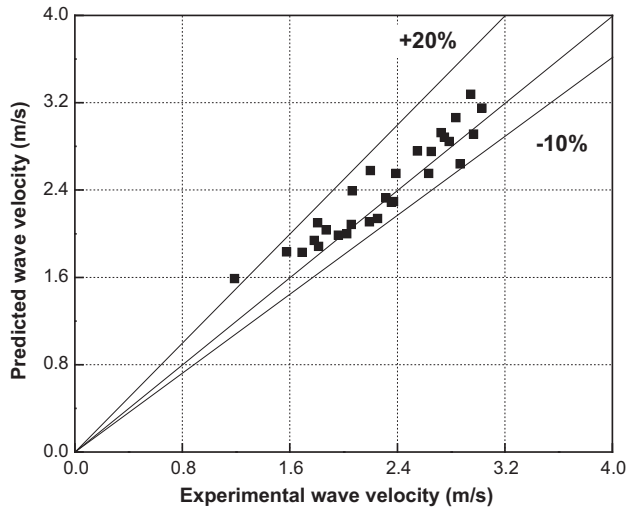


Fig. 19. Comparison of the present measurements of the wave velocity with the Kumar et al. [40] correlation.

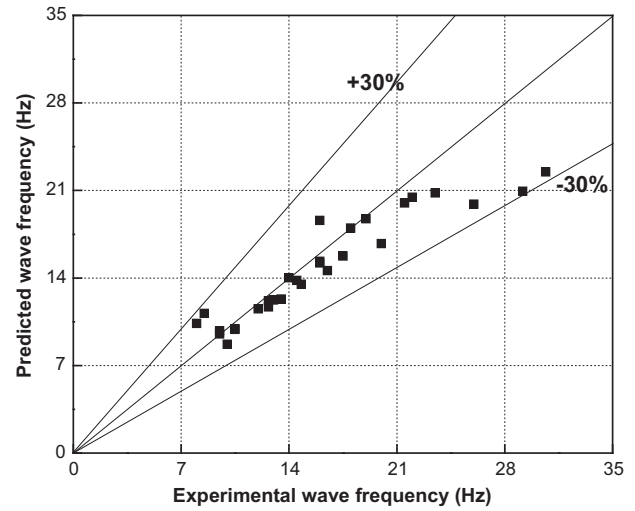


Fig. 21. Comparison the present measurements of the average disturbance wave frequency with the Sekoguchi et al. [41] correlation.

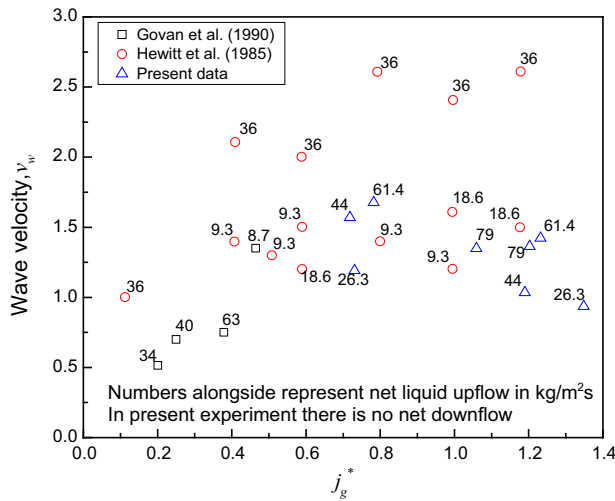


Fig. 20. Comparison of present flooding wave velocity with previous [3,1] data.

$$\psi = \frac{Re_l^{2.5}}{Fr_g} \quad (14)$$

$$Fr_g = \frac{j_g}{\sqrt{gd}} \quad (15)$$

Ψ is a non-dimensional parameter.

This correlation was tested by Hazuku et al. [42] for their experimental data in an 11 mm diameter test section. They found the correlation to predict their data within $\pm 25\%$. The Sekoguchi et al. [41] correlation when compared with the current dataset is seen (Fig. 21) to predict similarly to within $\sim 30\%$.

6.2.2. Flooding wave frequency in churn flow

The frequency is compared (Fig. 22) against the same dataset [3,1] as for flooding wave velocity. Frequency is seen to increase with gas and liquid flow. This observation is in agreement with previous data. However, for a given j_g^* and liquid flow rate, the presently obtained frequencies are less than previous values. This might be due to the inclusion of short lived waves by previous

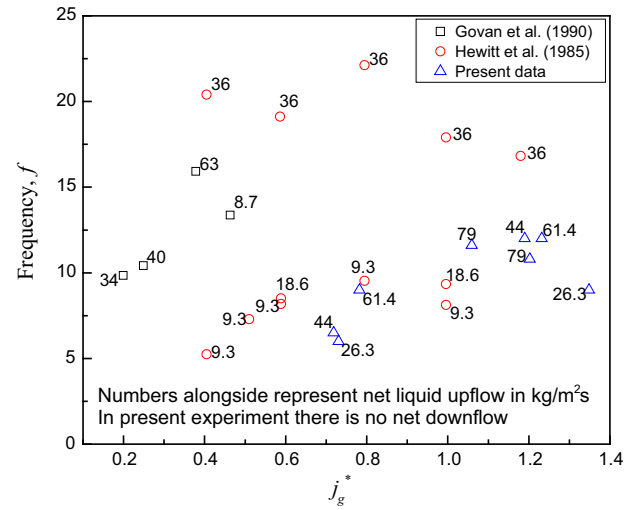


Fig. 22. Comparison of present flooding wave frequency with previous [3,1] data.

workers. In the present work only persistent flooding waves are considered.

6.3. Closing remarks

Given the good consistency of the earlier correlations with the present measurements, with respect to both disturbance wave frequency and speed, it seems safe to conclude that they can reasonably be applied, outside their region of origin, in the low-flow region annular regime explored in the present experiments.

The flooding wave velocity and trends obtained in the present experiments are similar to previous works. The frequency however is less possibly due to consideration of only sustained flooding waves in the present work.

7. Conclusions

A new set of data are presented for upward low pressure air-water flow for a 11 mm ID pipe at superficial air and water velocities of $j_g = 6.0$ – 28.0 m/s and $j_l = 0.02$ – 0.08 m/s. This operating parameter space falls in a zone where both air and water flow rates

are significantly lower than similar studies in the literature for small diameter tubes, and is a part of the parameter space important for nuclear reactor safety studies.

High speed photographic images were analysed to obtain wave parameters such as wave velocity, frequency and wavelength for different air and water flow rates in the churn and annular flow regimes. Experimental data has been generated for film flow rate and film thickness in annular flow.

The usual flow regimes of churn and annular flow, were clearly observed in the study. An additional intermediate, pre-annular regime was observed, characterised by momentarily stationary ripples. Each of these flow regimes could be very accurately identified by the motion of ripples in distance-time plots.

The measurements on disturbance waves in annular flow are mostly consistent with previous observations and in reasonable agreement where the measured conditions coincide. Importantly, the predictions in this parameter space made by earlier correlations mostly developed from measurements at higher flow rates were found to be reasonable, and this provides some validation for their employment under these conditions.

Additional quantification of the variation in time period between successive waves has been presented. The results on velocity of flooding waves in churn flow are also reported. It is seen that the velocity of flooding waves reduces with increase in gas flow. This behaviour is opposite to that observed for disturbance waves in annular flow.

The relative velocity ($j_g - v_w$) data generated in this work could be particularly useful for simulations of droplet deposition in annular flows. Present day correlations do not account for the effect of disturbance wave on deposition rates. In the light of the present work, the investigation of droplet deposition could usefully be revisited.

The transition from churn to annular flow has also been studied from the point of view of axial expanse of the disturbed region. An attempt has also been made to quantify the length of the disturbed region in terms of pipe diameters. It is seen that the disturbance length drops sharply in passing from churn to annular flow.

The experimental velocity and frequency data in the annular flow regime has been compared with Pearce [38], Kumar et al. [40] and Sekoguchi et al. [41] correlations. The results were found to be similar to what had been observed by previous investigators [16,42].

A comparison of the present and previous [3,1] velocity and frequency data on churn flow waves indicated a reasonable agreement for velocity. The lower frequency observed in present work is a result of the decision to included here only persistent waves.

In annular flow, the wavelength was found to be a strong function of film Reynolds number up to the critical film Reynolds number for onset of entrainment, $Re_{f,c}$. This observation coupled with visual observations led us to identify a physical relationship between $Re_{f,c}$ and the wavelength. This significance of $Re_{f,c}$ is a new and important conclusion of the present study.

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